INTERNATIONAL GCSE

Biology

9201/1

Paper 1

Mark Scheme

November 2023

Version: 1.0 Final

23BY92011/MS

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	a gene		1	AO1 3.5.3b 3.5.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	sex cells		1	AO1 3.5.3b 3.5.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3		(no mark for identifying cell)		AO2 3.5.3c 3.5.2c 3.5.2h
	(cell A nucleus) **–** skin cell because it has pairs of chromosomes	allow skin cell because it is diploid	1	
	(cell B nucleus) **–** sex cell because there is only 1 copy of each chromosome	allow sex cell because it is haploid	1	
		if no other marks awarded allow 1 mark for twice as many chromosomes in cell A (skin cell) nucleus **or** half the number of chromosomes in cell B (sex cell) nucleus		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	46		1	AO1 3.5.3d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	(male) XY		1	AO1 3.5.3d
	(female) XX		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	homozygous		1	AO1 3.5.3f 3.5.2b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7 **Mark with 01.8 and 01.9**	<table><tr><td></td><td>B</td><td>b</td></tr><tr><td>B</td><td>BB</td><td>Bb</td></tr><tr><td>b</td><td>Bb</td><td>bb</td></tr></table>	all correct = 2 marks 1 or 2 correct = 1 mark	2	AO2 3.5.3g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.8 **Mark with 01.7 and 01.9**	ring around all BB and Bb genotypes		1	AO2 3.5.3g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.9 **Mark with 01.7 and 01.8**	3 : 1	ratio must agree with student's genotypes if no answer to 01.7 allow 3 : 1	1	AO3 3.5.3g 6.3.6

Total			**12**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	B		1	AO3 3.2.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	balance		1	AO4 3.2.2b 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	increases		1	AO3 3.2.2b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	29 (cm^2)	allow 28 to 30 (cm^2)	1	AO2 3.2.2b 6.3.2 6.3.11

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	11.0 x 0.12 1.32 (cm^3)		1 1	AO2 3.2.2b 6.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	a leaf has 2 sides / surfaces **or** this method only measures SA of one side of the leaf		1	AO3 3.2.2b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	11 x 2 : 1.32 16.6666…. : 1 17 : 1		1 1 1	AO2 3.2.2b 6.3.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8		answers must be comparative		AO2 3.2.2d
	temperature is higher	allow it is hotter	1	
	wind speed is faster / greater	allow it is windier	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.9	the stomata are opened and closed		1	AO1 3.2.2e
	by the guard cells (which surround them)		1	

Total			**14**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	body temperature		1	AO1 3.4.2e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	by using automatic systems		1	AO1 3.4.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	any **one** from: • breathed (out) • in urine		1	AO1 3.4.3a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4		in either order		
	urea		1	AO1 3.4.3b
	ions	allow salt(s) allow named ion / salt	1	

Question	Answers	Mark	AO/ Spec. Ref
03.5	**Level 2:** Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.	3–4	AO1 3.4.3d AO1 3.4.3d AO1 3.4.3e
	Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.	1–2	
	No relevant content	0	
	Indicative content • blood is filtered • in the kidney / nephron • glucose / ions / urea / water are filtered • proteins are not filtered • because they are too large • all the glucose is reabsorbed • ions needed by the body are reabsorbed • water needed by the body is reabsorbed • control by ADH • urea is excreted in the urine • excess water and ions are excreted in the urine		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	any **two** from: • (excess) amino acids are deaminated to form ammonia / urea • poisons / toxins are detoxified / broken down for (safe excretion) • old blood cells are broken down (and the iron stored)		2	AO1 3.4.3c

Total		**11**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	any **one** from: • (same) temperature • (same) concentration of protease • (same) rinsing and blotting technique • (same) batch of eggs	allow same egg allow eggs from same animal	1	AO4 3.2.4b 6.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	(0.26 + 0.31 + 0.27) ÷ 3 0.28		1 1	AO2 3.2.4b 6.2.3 6.3.4 6.3.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	the highest activity is at pH 5	allow it works fastest at pH 5 allow the optimum is pH 5	1	AO3 3.2.4b. 6.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	any **one** from: • use smaller intervals between pH values • test at pH 4 and pH 6 • test more solutions around pH 5	allow use a different protein	1	AO4 3.2.4b 6.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.5	zero error	allow systematic error	1	AO4 3.2.4b 6.2.3 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.6	any **one** from: • adjust balance so it reads zero • subtract 0.02 g from every (mass) reading		1	AO4 3.2.4b 6.2.3 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.7	(molecule) amino acid(s) (reason) protein has been digested / broken down		1 1	AO2 3.2.4c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.8	If sensory neurones damaged impulses will not travel (effectively) from receptors to the CNS If relay neurones are damaged impulses will not travel (effectively) within the CNS If motor neurones are damaged impulses will not travel (effectively from CNS) to the effector / muscle	allow if sensory neurone is damaged the person cannot sense stimuli **or** if sensory neurone is damaged the person cannot see the object	1 1 1	AO2 3.4.1b, c, d

Total		**12**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	all the populations of living things in the forest		1	AO1 3.3 3.3.3d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	processes that remove materials are balanced by processes that return materials		1	AO1 3.3.3d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3		ignore references to oxygen		AO2 3.3.3b
	The temperature is higher	allow it is warmer	1	
	there is (more) water / moisture		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	microorganisms	allow decomposers / bacteria / fungi	1	AO1 3.3.3b
	break down / digest the (dead) leaves / plants		1	

Question	Answers	Mark	AO/ Spec. Ref
05.5	**Level 2:** Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.	4–6	AO1 3.2.1a,d,e 3.2.2a 3.3.3c,d
	Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.	1–3	AO1 3.2.1a,d,e 3.2.2a 3.3.3c,d
	No relevant content	0	
	Indicative content • microorganisms respire • releasing carbon dioxide • which is taken in by new plants for photosynthesis • to make glucose (needed to grow) • glucose is needed to make lipids / proteins / other named molecules • glucose is needed for respiration • to release energy for growth • mineral ions are released into the soil • proteins / amino acids are broken down • so nitrate ions are released into the soil • taken up by the roots of plants • used to make amino acids / protein For **Level 2** reference to at least 2 substances and how they are used for growth must be made.		

Total		**12**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	a vector is used to insert the gene into the required cells the required gene is cut from a chromosome using an enzyme		1 1	AO1 3.5.5b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	Increase = 17 17/7 x 100 = 242.857……. 243 (%)		1 1 1	AO2 3.5.5d 6.3.3 6.3.4 6.3.11

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	chemical X is less because it has been used therefore the amino acids are still being made		1 1	AO3 3.5.5d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	any **three** from: • increased / high crop yield • more food production • farmers can make more money / profit • no / less need to remove weeds by hand	answers must be comparative allow reference to less starvation or better food security in countries which rely on cassava	3	AO3 3.5.5d,e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	not known if it affects wild flowers / insects		1	AO1 3.5.5f
	not known if it affects human health (when eaten)		1	

Total		**12**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	any **two** from: • eat less fatty foods • eat more fruit and vegetables • lose weight if obese / overweight • take regular exercise	allow maintain a BMI below 25 allow eat less salt allow do not smoke allow drink less / no alcohol allow take statins to reduce blood cholesterol	2	AO3 3.2.3e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	blood flow in (coronary) artery is reduced		1	AO2 3.2.3e 3.2.6b,f,g
	(so) not enough oxygen reaches the heart muscle (through blocked coronary arteries)		1	
	(so) respiration will not be occurring at a high enough rate		1	
	to release sufficient energy	do **not** accept energy produced / made / created	1	
	for the heart to contract / pump fast enough		1	
	to supply oxygen quickly enough for (exercising) muscles **or** so breathing rate increases to try to supply more oxygen (so they become breathless)		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	any **one** from: • so that it can easily pass through blood vessels • so it can be stretched open • so it can expand and contract with the artery	allow so it does not damage / tear blood vessels allow so the balloon can inflate	1	AO3 3.2.3e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4	to open up the artery / stent	allow to squash / flatten the fatty material allow to push the stent against the artery wall ignore to remove the fatty material	1	AO3 3.2.3e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	the coronary arteries of people vary in diameter	allow so the stent will fit securely / properly	1	AO3 3.2.3e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.6	because the (B) patient has anti–a antibodies (in their blood)		1	AO2 3.2.3t
	which will attack the A antigens on the (AB) donor red blood cells (and cause a reaction)	allow will bind / react with the A antigens on the (AB) donor red blood cells (and cause a reaction)	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.7	a protein		1	AO1 3.2.3r

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.8	the red blood cells could then be given to anyone (in a transfusion) whatever their blood group		1	AO3 3.2.3r,s,t
	because they would not attack / bind / react with the antibodies (in the patient)	allow because they would not be clumped / agglutinated / destroyed	1	AO2 3.2.3r,s,t

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.9	enzyme	allow protease	1	AO2 3.2.4b

Total		**17**